

Lecture 4: Experimental Design

CMSC/DATA 320: Introduction to Data Science

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Announcements

- PA1, WA1 due this Sunday
- Office hour schedule:
 - Anna on Zoom: Tue 11 am-12 pm
 - TAs in person: Tue 12-1 pm, Fri 9-10 am, Fri 4-5 pm
 - TAs on Zoom: Thu 11 am-12pm
- In-person office hour location: **IRB 2216**
- Quiz grades and solutions will typically be available a week after the quiz
- Quiz make-ups now also available after Wednesday's class

Experimental Design

Experimental Design

A systematic method for planning research to establish cause-and-effect relationships. Includes deciding how to collect your data.



R. A. Fisher (1890-1962)
Founding father of Modern Statistics
Geneticist



Experimental Design

Experimental design is the process of planning, carrying out, and analyzing experiments to test a hypothesis.



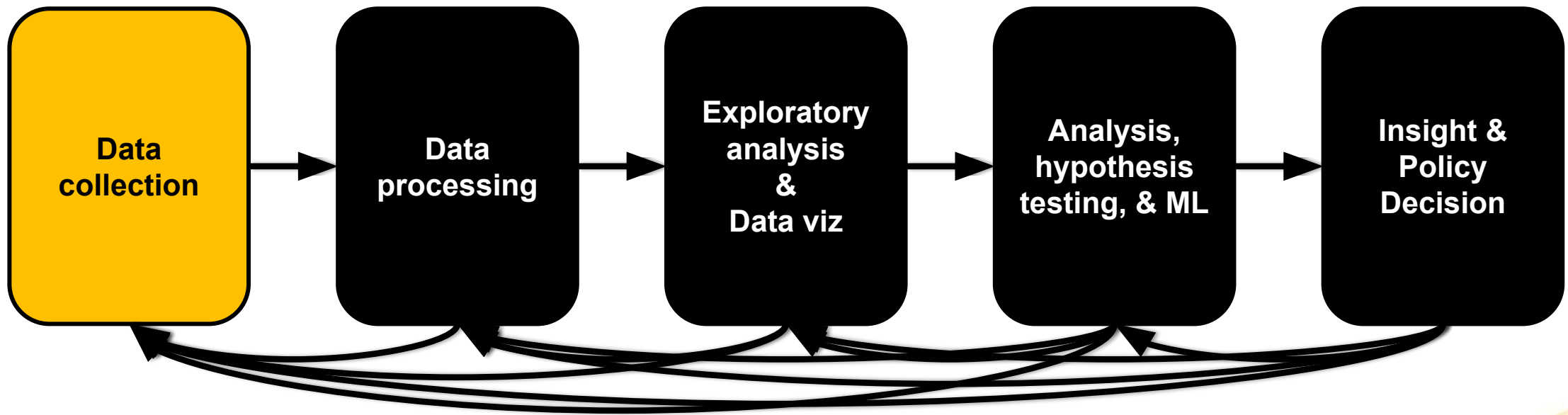
Let's see if the subject responds to magnetic stimuli... ADMINISTER THE MAGNET!

Interesting...there seems to be a significant decrease in heart rate. The fish must sense the magnetic field.

From: <http://www.hawaii.edu/fishlab/NearsideFrame.htm>

Experimental Design in Data Science

Experimental design is a crucial aspect of data science that focuses on **planning** and **conducting experiments** to **gather meaningful data**.



Experimental Design Steps in a data science project

- ❑ **Define the Problem** or Research Question to address
- ❑ **Identify the variable(s)** including potential confounding variables and the population/sample you will study
- ❑ **Formulate a Hypothesis**
- ❑ **Develop a detailed plan** for data collection, ensuring representativeness when using a sample
- ❑ **Select Data Collection Methods**
- ❑ **Collect Data**

Then continue on to the next Data Science Life Cycle step...



Define the Problem or Research Question

What will the weather be like for the next 10 days?

Which courses are likely to have the highest demand next semester?

How many visitors will the website named "X" receive over the next week?



Which loan applicants are likely to default in the next year?

Which students are at risk of dropping out of an online course?

Will flights be delayed over the next 10 days?

Will upcoming movie "Avatar 6" be a box office hit or miss?

Which subscribers are likely to cancel their Netflix subscription soon?

Asking the right questions before solving a DS problem is a great start! And be specific!

Experimental Design Topics

- **A perfect scenario for an Experiment**
 - Identifying Variables & Population / Sample of the Study
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 - Methods for Collecting Data
- 

Example: Online Retail

Let's say you're a data scientist working for an online retailer, and you want to test **whether changing the color** of the "Buy Now" button on your website affects the click-through rate (CTR)

- Click-through rate (CTR) = the percentage of users who click on the ad after seeing it

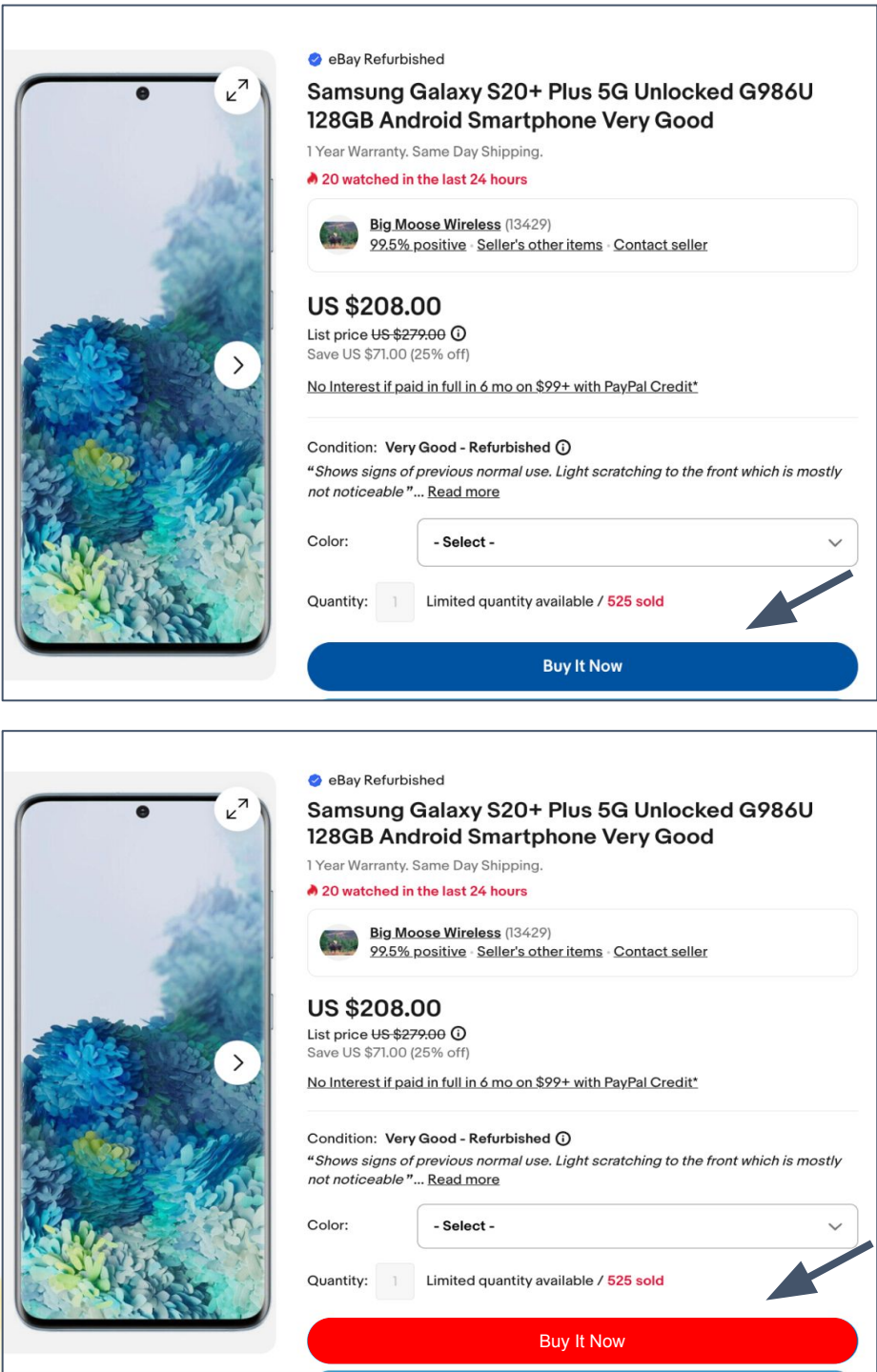
What is your problem definition?

Which version of the “Buy Now” button (Option A (default) or Option B (red)) is **more likely to maximize** the CTR

What is you Optimization Criteria? What do we want to maximize?

CTR → we want to select the ad options with button “buy now” that leads to **Higher CTR**

Ques: How can we set up an experiment to collect data in this case?

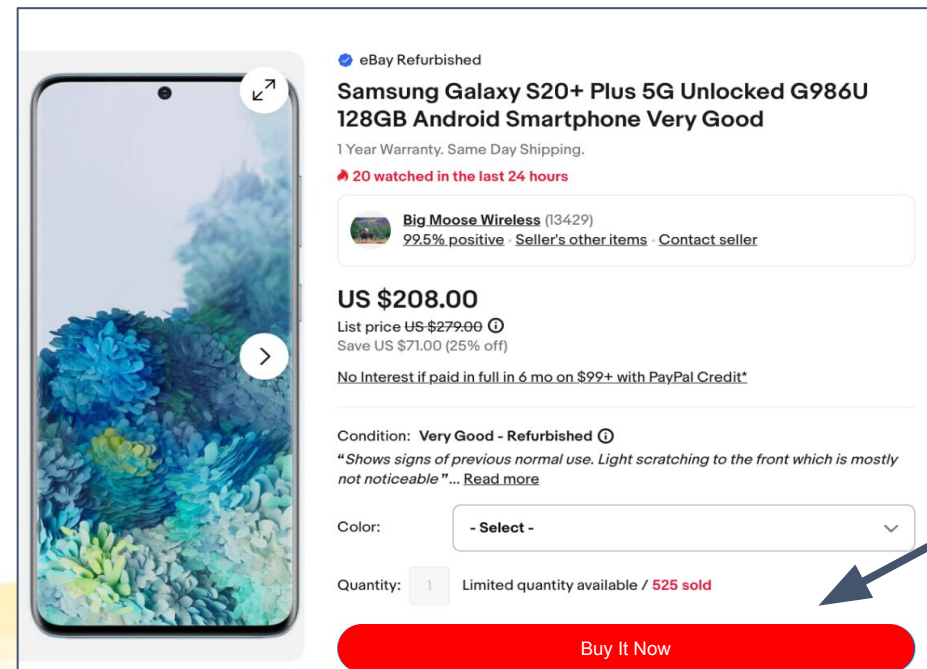
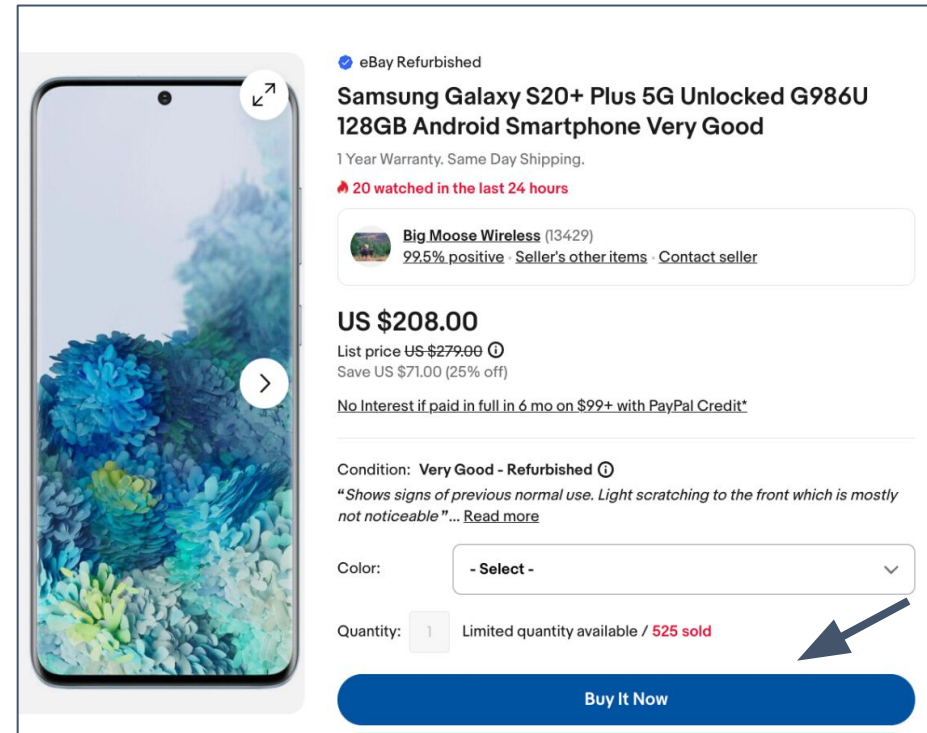


Que: How can we set up an experiment to collect data?

Data Size / Sample: No. of website visitors

- Views the original website with **existing button color**.
- Experiences no changes; baseline for comparison.
- Views the same website but with a **different color for the "Buy Now" button**.
- Experiences the change you want to test.

- **Collect data** on the click-through rates for both groups over a specific period.
- **Compare** the click-through rates (CTR) between the groups **after** the experiment
- **Check** if there's a significant difference [LATER TOPIC IN THIS COURSE], we can infer that the **change in button color influenced the click-through rate**.



Ques: How can we set up an experiment to collect data in this case?

Data Size / Sample: No. of website visitors

- Views the original website with **existing button color**.
- Group experiences no changes

CONTROL GROUP

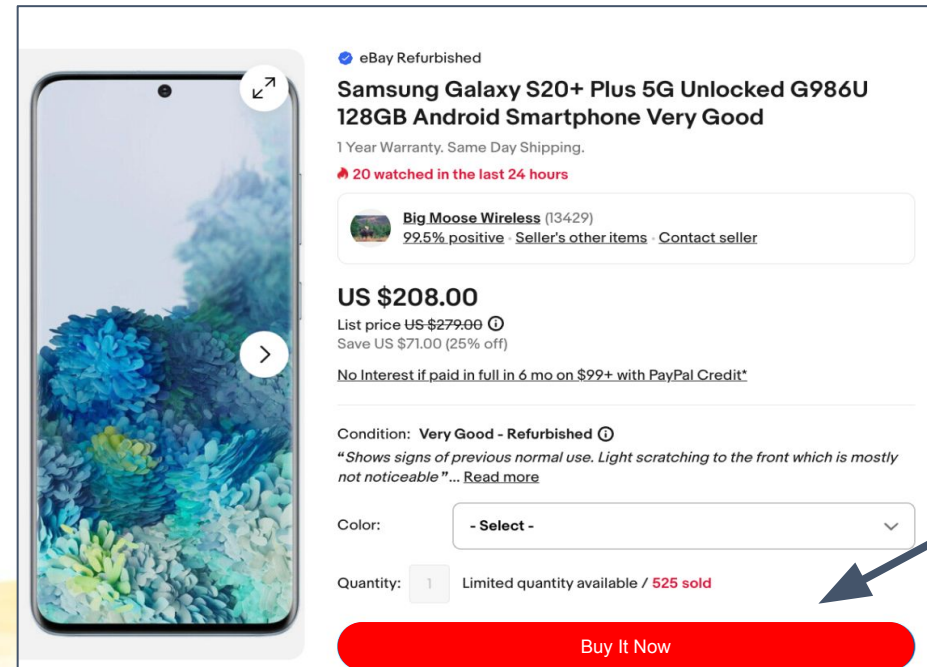
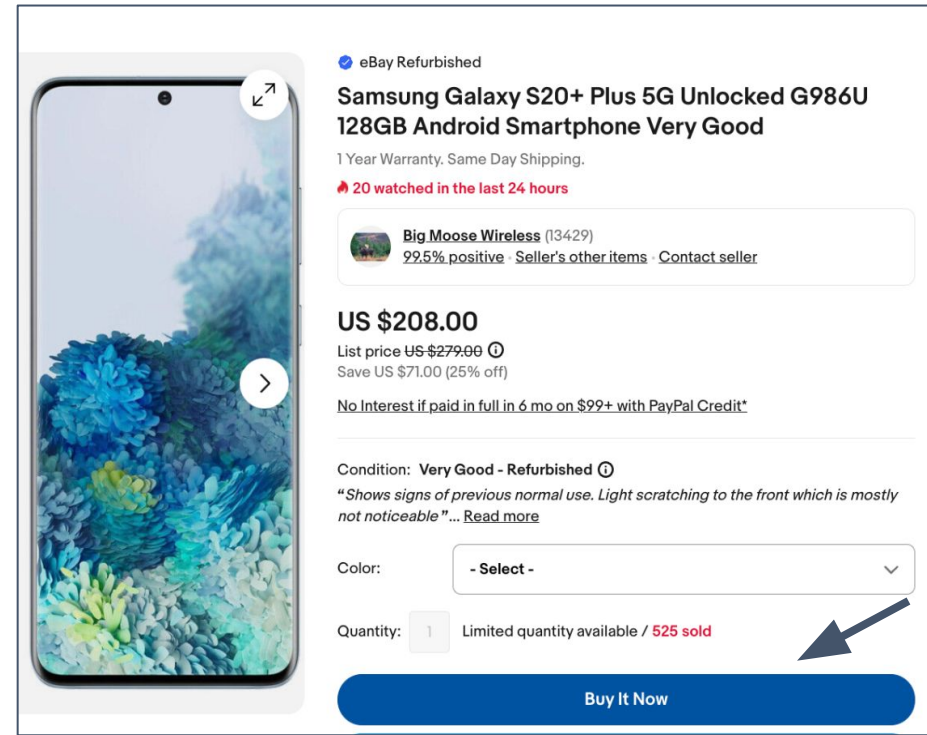
- Views the same website but with a **different color for the "Buy Now" button**.
- Group experiences the change you want to test.

TREATMENT GROUP

Ques: What are the variables here?

- Click-through rate (what we measure) ← Dependent Variable
- Color of the "Buy Now" (what we manipulate) ← Independent Variable

The 2 groups need to be "the same" in everything but the button color. Random assignment makes that happen.



Experimental Design Topics

- A perfect scenario for an Experiment
 - **Identifying Variables & Population / Sample of the Study**
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 - Methods for Collecting Data
- 

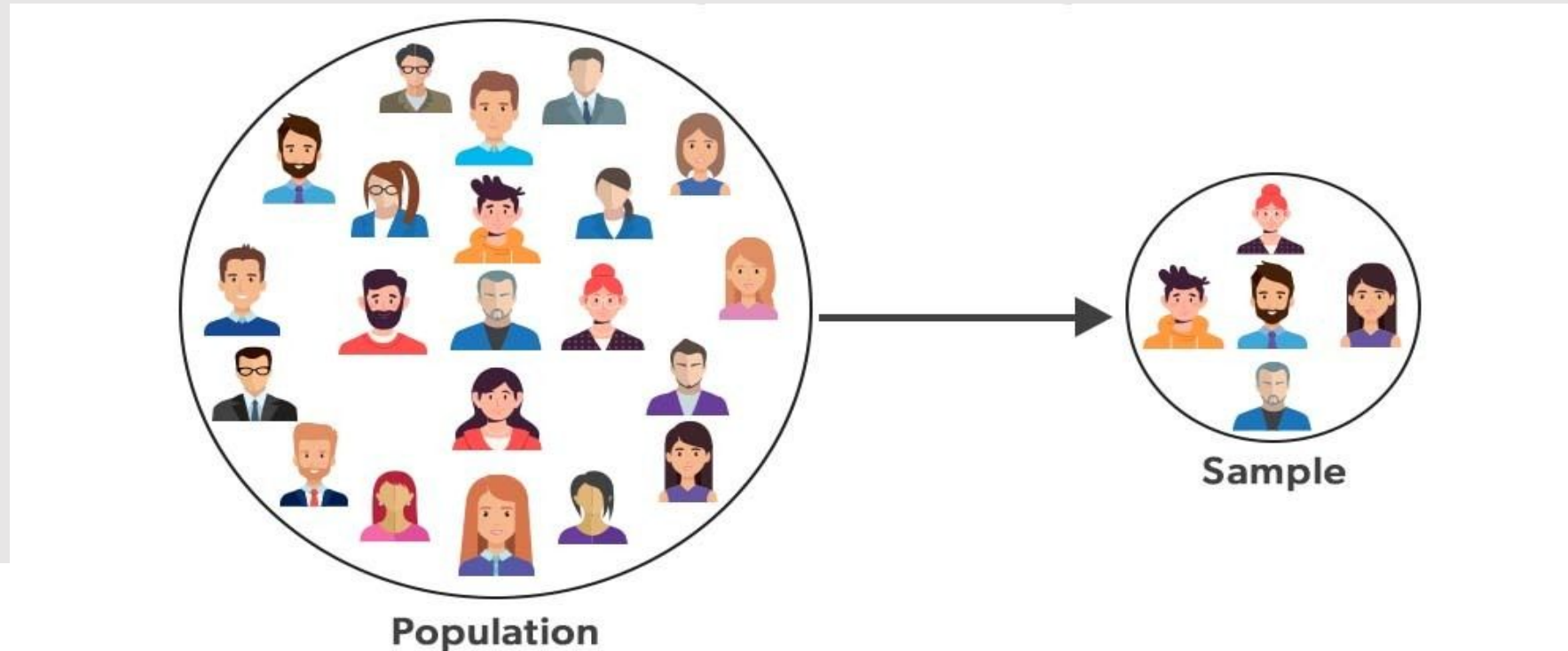
Identify variable(s) of interest & population of the study

Once the problem is defined, identify the **variable(s)** of interest that are relevant to your research question.

Also, specify the **population** or **sample** that your study will focus on.



Terminology: Population and Sample



Population: The entire group that we want to learn about.

Sample: A smaller group from the population that we actually study to draw conclusions about the entire population.

Terminology: Variable

A variable is *any characteristic* that is recorded for subjects in a study.

Example: “gender,” “major,” “age,” and “GPA” might be variables for a study about college students.



Identify variable(s) of interest & population of the study

2 types

- **Independent variable (IV)**
 - Factor(s)/cause(s) that may influence the outcome
 - We can manipulate or change to see how it affects the dependent variable
- **Dependent variable (DV)**
 - Outcome of interest (what we measure)
 - expected to change as a result of changes to the independent variable

Important: While independent variables produce dependent variables, it is also true that **multiple independent variables may influence one another.**

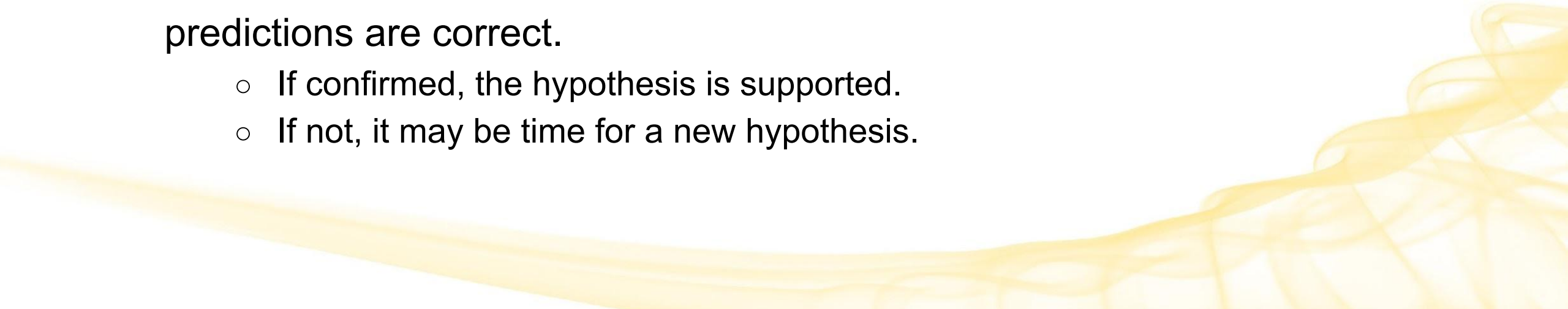
Experimental Design Topics

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Come up with a Hypothesis

Hypothesis of your experiment is the statement you want to test.

ex. “If hypothesis X is right, then Y should be true...”

- A testable explanation/statement that addresses the question.
 - Represents an educated guess as to the relationship between variables and the outcome of the experiment
 - Experiments or Observations are conducted to see whether the predictions are correct.
 - If confirmed, the hypothesis is supported.
 - If not, it may be time for a new hypothesis.
- 

Example Hypothesis

"If the **color of the button** (____ variable?) is changed from Blue to Red, then **Click-Through Rate** (____ variable?) will also increase."

Remember:

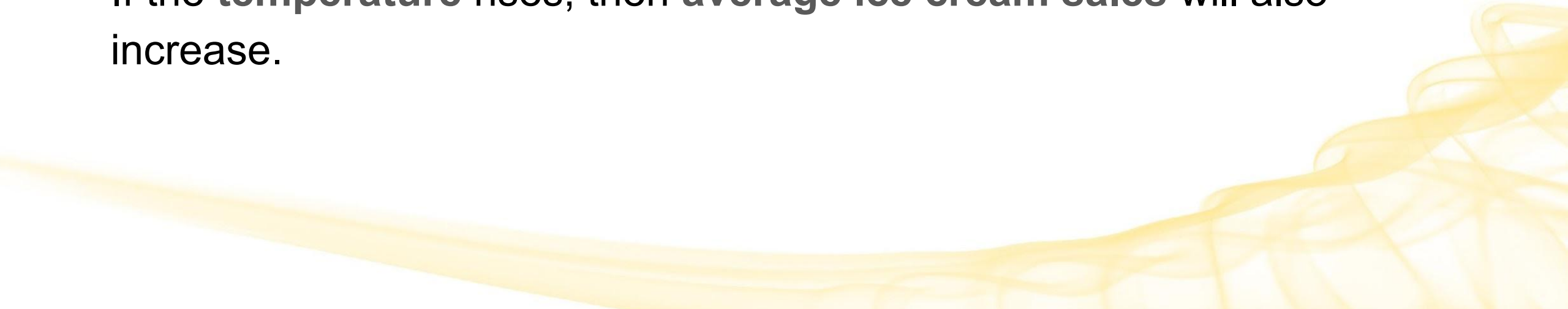
- **Independent variable (IV):** factors that may influence the outcome
- **Dependent variable (DV):** outcome of interest

More Hypothesis Examples

As **books read** increases, **average literacy** also increases.

If **exercise duration** is extended, then **average calories burned** will also increase.

If the **temperature** rises, then **average ice cream sales** will also increase.



Experimental Design Topics

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Select a Data Collection Method and Collect Data to Test your Hypothesis

Before collecting data to test your hypothesis, you first have to consider the problems that can cause errors in your result, one of them being a **confounder**.

Analyze if any other variables (**beyond the ones you intended to manipulate or measure**) could have impacted / influenced your results

Note: if true, we may need to redesign the experiment to ensure more accurate and reliable conclusions.

What is a confounder?

An extraneous variable that **may affect** the relationships between dependent and independent variables.

- **May not be the focus** of the study but **can impact the results** by influencing the dependent variable.
- An additional factor that is not controlled for but could potentially influence the outcomes.

A confounder can lead to incorrect conclusions about the true effect of the independent variable on the dependent variable.

Back to our Example Hypothesis

factor that is manipulated or changed;
expected to have an impact on the
dependent variable

"If the amount of **study time** (independent variable) is increased, then **exam scores** (dependent variable) will also increase."

the outcome or response that is measured in the study; expected to change based on variations in the independent variable.

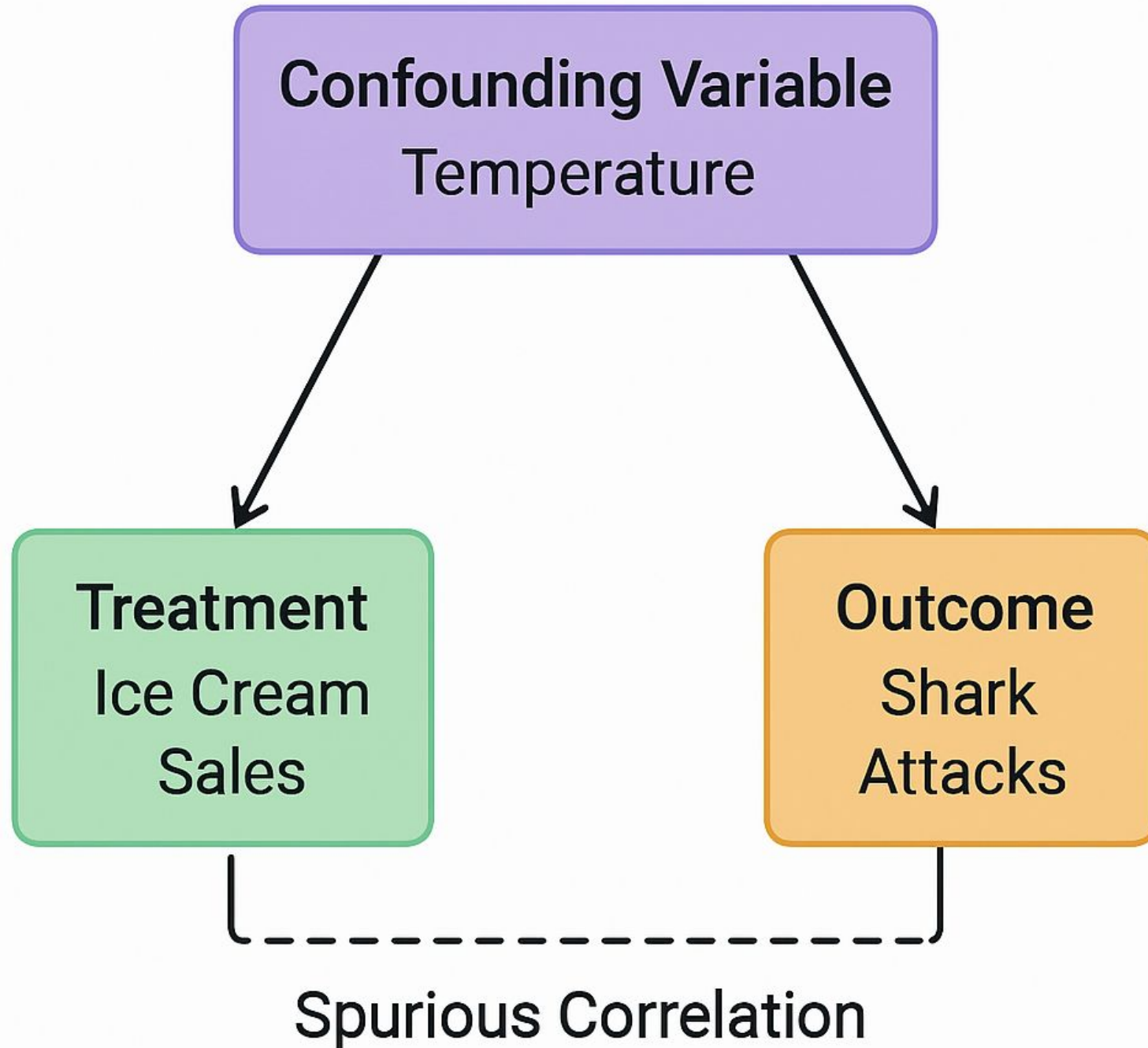
What is a potential confounding variable?

Confounding Variable
Temperature

Treatment
Ice Cream
Sales

Outcome
Shark
Attacks

Spurious Correlation



Example Experimental Design Flow: Polling

1. **Problem Formulation:** Imagine you're a data scientist tasked with predicting the outcome of a political election using a dataset of voter preferences. Your goal is to **design an experiment that accurately represents the entire population's voting behavior.**

Questions:

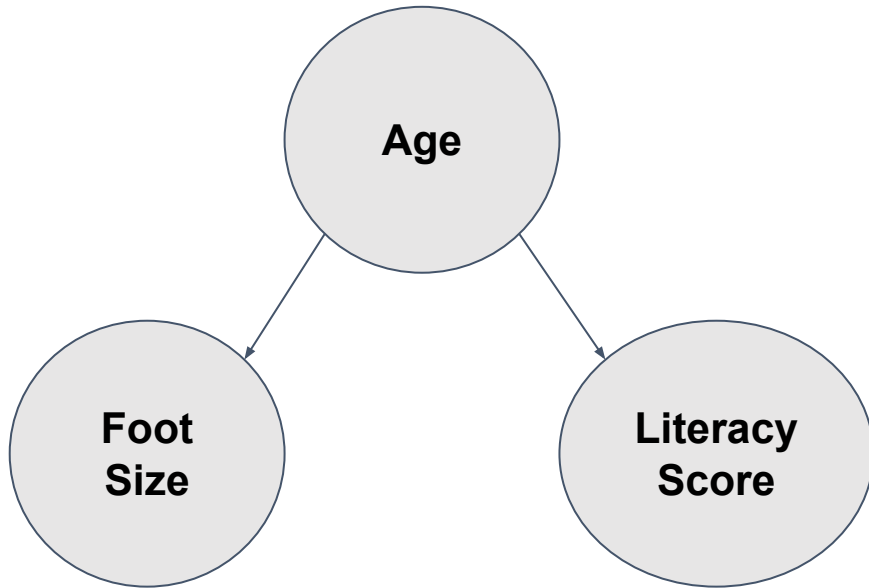
- How do we know which candidate is ahead?
- Can we create the IDEAL POLLING?

Eliminate confounding variables as much as possible → Sample Bias, geographic representation, Population Proportion Bias, Demographic Mismatch and many more to make the data as accurate as feasible.

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Motivating example: dealing with a confounder



- Foot size and literacy score are positively correlated
- Larger foot size is not the cause of a higher literacy score
 - They're independent (more on this later)
- Age is a **confounder**
 - If you know foot size, it hints at age, which hints at literacy score
- By **controlling** for age (the confounder), we can avoid foot size $\leftrightarrow$ literacy problem
 - Consider at each age group separately
 - Or add Age as another Independent variable

Dealing with Confounders

Considering and addressing confounders in experimental design is crucial to ensure accurate results, preventing external factors from influencing outcomes and ensuring the correct design of experiments.

There are three ways you can deal with confounders:

- Control
 - Randomization
 - Replication
- 

Technique 1: Control

Manage or eliminate the **impact** of confounding variables

- Make sure to collect the confounding variable
- Make the confounding variable part of the analysis (e.g. in regression), or compare things only when the confounding variable values are equal
 - E.g., does temperature correlate with ice cream sales when it is **a holiday** (potential confounder)?

Or: Check the effect of “age”

"As **books read** increases, **literacy** also increases."

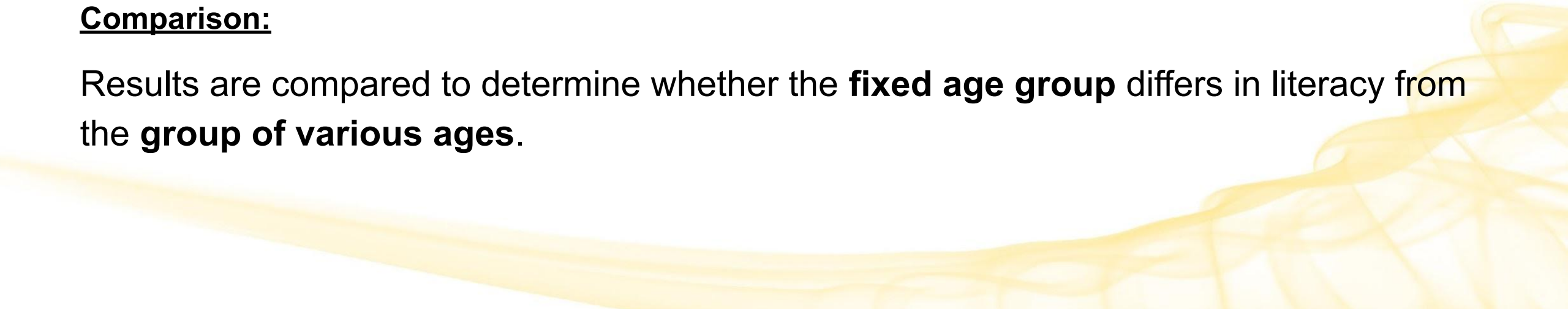
✓ We can measure the age of each individual to see the effects of age on literacy.

Experimental Design:

- Control group → participants of fixed ages.
- Treatment group → participants of the range of ages.

Comparison:

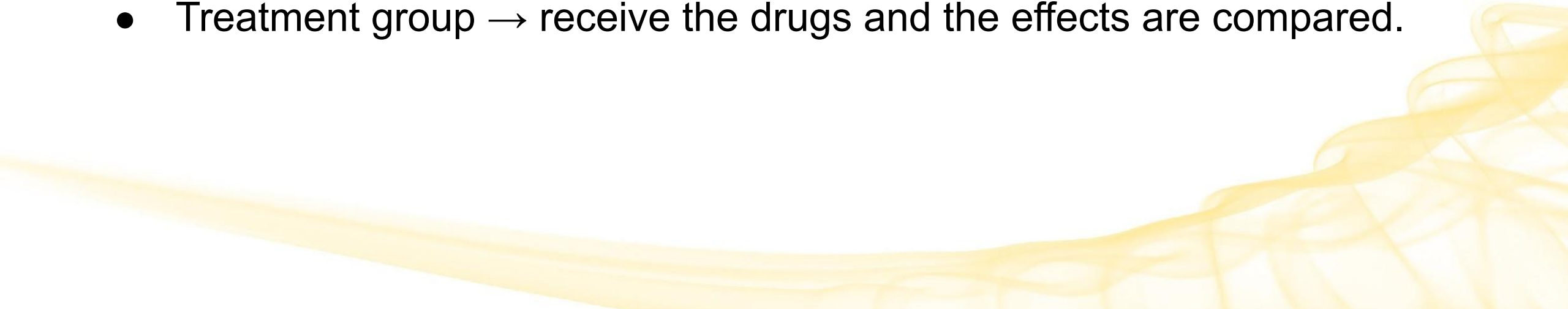
Results are compared to determine whether the **fixed age group** differs in literacy from the **group of various ages**.



Another general example: a classic experimental design for drug testing

We want to **test the effect of a drug** on patients.

Experimental Design:

- Control group → won't receive the drugs (used a baseline - not exposed to the independent variable)
 - Treatment group → receive the drugs and the effects are compared.
- 

Technique 2: Randomization

Helps make groups fair by deciding things **randomly**.

Randomly assign participants to

- The control group (**won't receive the drugs**) or
- The treatment group (**receive the drugs**)

Why? To minimize systematic confounding, reduce the risk of bias in either group being enriched for confounders, and help distribute confounding variables equally

- ❑ This minimizes the likelihood of systematic confounding
 - Reduces the risk of bias in either group being enriched for confounders.
 - Each participant is assigned to one of the two groups, not both simultaneously.

Technique 3: Replication

Doing your experiment **more than once** to be more certain.

- Repeat the study multiple times to assess the consistency and reliability of results, helping to identify and account for potential confounding factors.

If replication is done (with a new set of data) and it **produces the same conclusions**, this shows that the experiment is strong and has a good design and suggests that **confounders are less likely to be influencing** the outcomes.

Experimental Design Topics

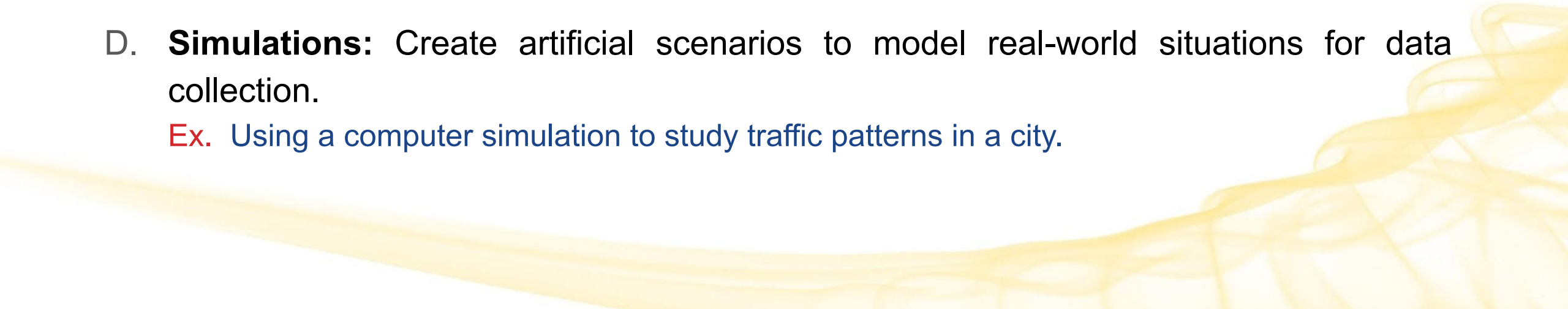
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 - **Methods for Collecting Data**
- 

Methods for Collecting Data

Below are some common methods/way to collect data if pre-existing datasets are not available.

- Observational studies
 - Surveys
 - Experiments
 - Simulations
- 

Methods for Collecting Data

- A. **Observational studies:** Observe and record data (variables) without intervening or manipulating variables (observe, don't change anything intentionally).
Ex. Observing tiger behavior in their natural habitat without any external influence.
 - B. **Surveys:** Collect information through structured questionnaires or interviews.
Ex. Conducting a survey to gather opinions on a political issue.
 - C. **Experiments:** We actively change something to see what happens.
 - D. **Simulations:** Create artificial scenarios to model real-world situations for data collection.
Ex. Using a computer simulation to study traffic patterns in a city.
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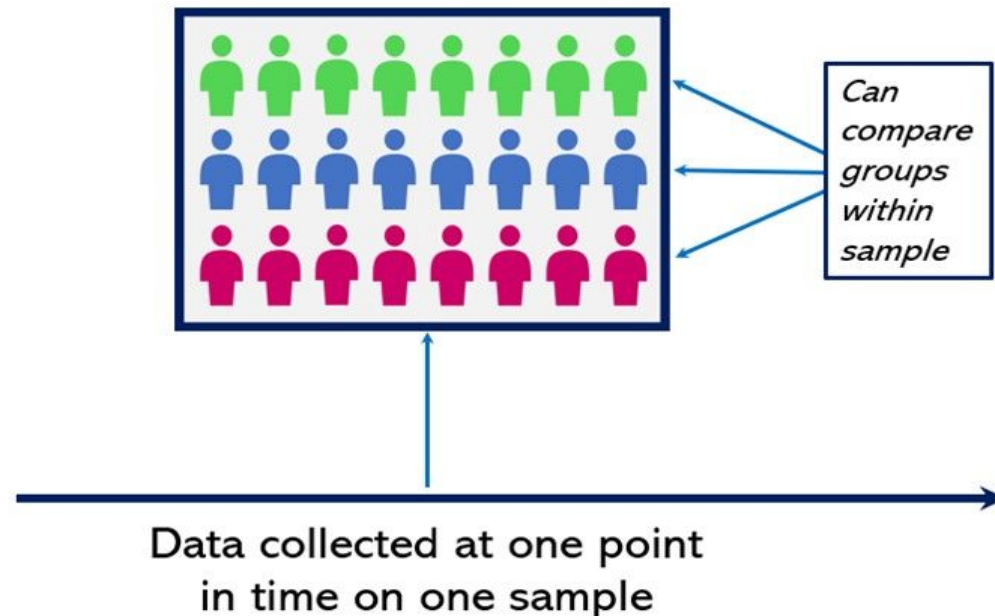
A. **Observational Studies**

- Cross-sectional studies
 - Retrospective (case control) studies
 - Prospective (longitudinal or cohort) studies
- 

A. Observational Studies

Cross-sectional studies: Collects data from many different individuals at one specific single point of time

- Taking a picture of a group **right now** to see what they're like.
- Surveying individuals of various age groups **on a single day**



A. Observational Studies

Retrospective (case control) studies: Looking backwards at studies of events in the past to examine the relationship between the exposure and the outcome.

Ex. Investigating past events to find out what might have caused something.

- **Interviewing severe dengue patients** to ask about their experiences and medical history, studying disease spread dynamics.
- **Studying past smoking history in lung cancer patients** (cases) and non-cancer individuals (controls) to identify associations and understand their impact on health outcomes.

A. Observational Studies

Prospective (longitudinal or cohort) studies: Researchers follow and observe a group(s) of people (called a **cohort**) closely **over a period of time**, collecting data on their exposure to a factor of interest (observe changes; identify potential causes).

Observing a cohort over time to see how things change

Ex. tracking their smoking habits and health outcomes to determine the association between smoking and the development of respiratory diseases.

- But! Sometimes people stop participating, which can make things a bit tricky to understand.
- These tend to have high dropout rates



B. **Surveys** (a specific type of observational study)

A survey is used to investigate characteristics of a population.

- Frequently used when the subjects are people, and questions can be asked of them.
 - When designing a survey, you must be very careful of wording (and sometimes ordering) of the questions so that the results are **not biased**.
- 

C. Experiment

In an experiment, a researcher assigns a **treatment** and observes the response.

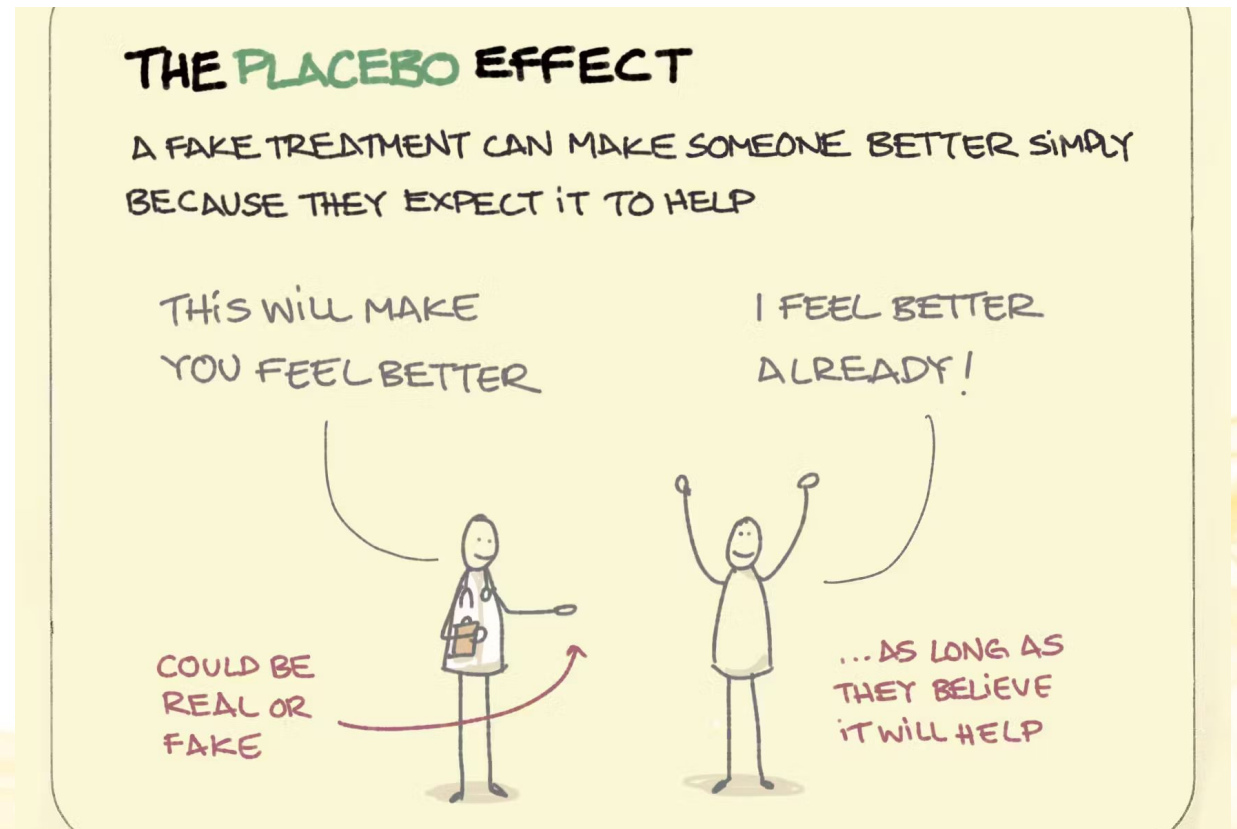
- Observe effects on subjects after the application of some treatment
 - Might want to compare a treatment versus a control or multiple treatments
receives intervention/actual treatment receives no intervention or receives a placebo (fake treatment)
- The more variables you can control for, the better your experiment

Placebo effect

Placebo Effect: Belief in treatment leads to feeling better even without actual treatment.

- A person believes psychologically that a certain treatment is positively affecting them, even though no treatment or ineffective treatment was given.

Ex. When a person's physical or mental health appears to improve **after taking a placebo or 'dummy' treatment.**

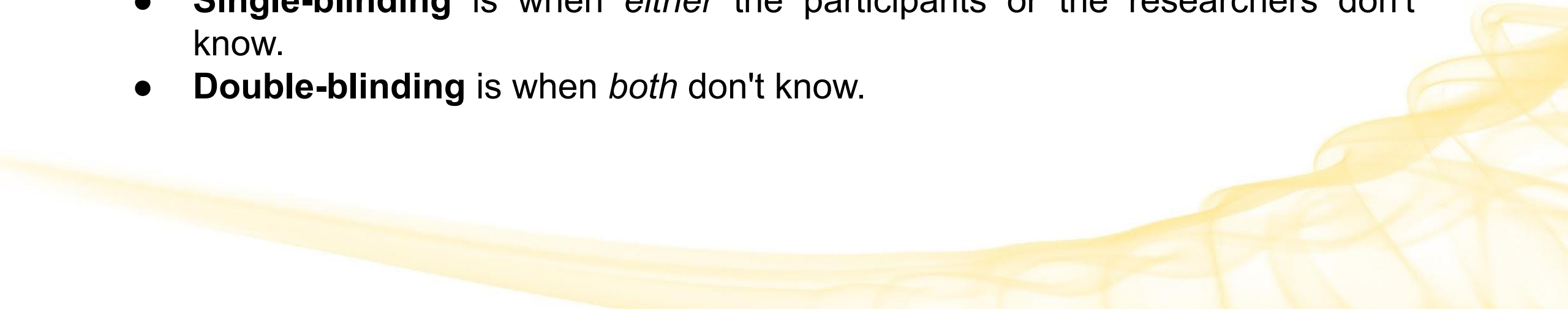


Blinding

A common technique to minimize bias due to the placebo effect in Experimental Design.

When the people involved in an experiment **don't know who's getting the real treatment and who's not.**

Prevent Biases. It is a method used to make the subjects “blind” to which treatment (or placebo) they are being given

- **Single-blinding** is when *either* the participants or the researchers don't know.
 - **Double-blinding** is when *both* don't know.
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D. Simulation

A simulation uses a mathematical, physical, or computer model to replicate the conditions of a process or situation.

- Frequently used when the actual situation is too expensive, dangerous, or impractical to replicate in real life

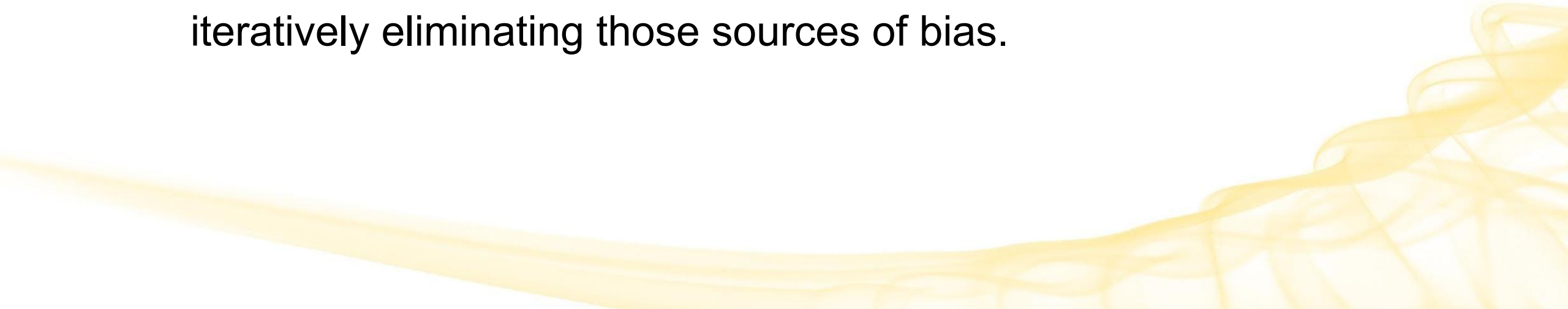


The Fundamental Rule of Data Collection

Your data must be **representative** of the population you want to study.



Keep in mind that

1. It is almost impossible to be certain that your experiment has completely removed all forms of bias.
 2. It is necessary to consider possible sources of bias and **highlight them in your analysis**.
 3. Ideally, future experiments would improve upon your method by iteratively eliminating those sources of bias.
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Quick Class Task

Identify which method for collecting data (observational study, an experiment, a simulation, or a survey) is **best** in each of the following situations and **explain your answer**.

1. The effect of a severe earthquake would have on the Salt Lake Valley.
2. Whether or not a certain coupon attached to the outside of a catalog makes recipients more likely to order products from a mail-order company.
3. Whether or not smoking has an effect on coronary heart disease.
4. Determining the average household income of homes in Salt Lake City.